DIFFERENTIAL EMISSION MEASURE INVERSION FOR SDO/AIA WITH DEEP LEARNING

SUMMARY AND WEB VISUALIZATION

1 INTRODUCTION AND APPROACH

The Differential Emission Measure (DEM) is one of the key diagnostics for the study of the thermal structure of the Sun. Obtaining the DEM often depends on optimization or sampling based methods that are computationally expensive. Our objective is to build a system that works with full SDO/AIA resolution and produces a DEM output, orders of magnitude faster than the classical inversion methods. Therefore, we have developed and evaluated deep learning-based systems that can accurately emulate the outputs of these methods.

Once these systems are trained, they can accept a calibrated AIA optically-thin EUV image stack and produce a DEM profile along with uncertainties per-DEM bin. Each model is trained to emulate a specific DEM solver. Among the various DEM inversion approaches in the literature, including Markov Chain Monte Carlo methods [1], regularized least squares [2], maximum likelihood approaches [3], and hybrid iterative techniques [4, 5], we selected two complementary methods based on their distinct optimization formulation and widespread adoption in the community. We have therefore developed reference Python implementations of the Cheung et al. [6] basis-pursuit sparse linear program approach and the Athiray and Winebarger [7] Elastic Net approach ("*Basis Pursuit: BP*" and "*Elastic Net: ENet*", respectively). The BP method employs pure ℓ_1 regularization, which promotes sparsity in the DEM solution. Although the ℓ_1 regularization does not follow a particular physical principle, it avoids overfitting and ensures positivity. The ENet method combines ℓ_1 and ℓ_2 regularization, which takes on the assumption of smoothness throughout the DEM bins. These approaches allow us to capture two specific physical scenarios, that are computational tractable and easily comparable. We include mathematical details in Appendix A.

We have additionally built methods that use Hinode/XRT data to constrain high-temperature bins (while not requiring XRT at inference time, meaning the time at which the model is queried for a new DEM of a new target), and we have built the ability to reconstruct multiple solutions based on sampling simulated photon noise in order to identify uncertainty. The Neural Networks operate on each pixel at the original AIA resolution and are based on modernizing the DeepEM approach originally proposed by Wright et al. [8, 9] on three fronts. *1) Inputs:* rather than directly use the per-pixel EUV observations, our approach first lifts the data into a high-dimensional space that enables fine-grained distinctions to be made more easily; this follows work in neural radiance fields [10]. *2) Network body:* we substantially expand the network width and depth to increase its learning capacity, and pick more modern nonlinearities (i.e., SiLU, which is differentiable everywhere). *3) Network output/uncertainties:* we predict both a single *best* estimated output as well as a distribution over possible outputs (generated by running a given solver on a range of perturbed inputs). The distribution prediction system follows our earlier work from SuperSynthIA [11], in which distributions are represented via a binning of the space. By breaking the output into these bins, the model can infer a distribution of probabilities over values (e.g., 40% chance of 0), which allows for more complex uncertainty estimates.

We have trained multiple of these methods on a 2 year dataset at twice-daily cadence to match XRT and allow for optional high-temperature bin constraints. To generate output inversions, we run our Python implementation on the full dataset with different configurations (e.g., BP vs ENet, AIA only or AIA+XRT). We highlight four models which we present here along a web visualizer. The four models (with abbreviations for the web visualizer) are:

- **NN on BP with AIA**: *Neural Network trained on Linear Program using only AIA*
- **NN on BP with AIA+XRT**: *Neural Network trained on Linear Program using both AIA and XRT*

- **NN on ENet with AIA**: *Neural Network trained on Elastic Net using only AIA*
- **NN on ENet with AIA+XRT**: *Neural Network trained on Elastic Net using both AIA and XRT*

2 WEB VISUALIZER

In the attached web visualizer[1], we present a diverse set of visualizations to compare the models. The visualizer shows in the top a set of three different drop-down bars to select a *model*, *a timestamp* and a visualization *mode*.

Use: Select a model, a timestamp and a mode in the drop-down bars to see the corresponding visualizations. Each visualization square can be zoomed in and out by scrolling the mouse wheel, with all pannels being synchronized. You can also pan around by clicking and dragging the mouse. Most importantly, the original full resolution images are accesible by doing right-click and opening the image in a new tab.

Preliminary notes: Our methods seem to closely emulate the reference methods. As was noted in the original DeepEM method, the neural net also smooths out some noise in the per-pixel Cheung et al. outputs. This smoothing behavior occcurs because the neural net model was fit on the the full training dataset simultaneously, unlike the Cheung et al. fit, which is done per-pixel. The network captures the multi-thermal structure recovered by Cheung et al., and is able to produce a confidence interval in a single pass (without any required perturbation or additional realizations at inference time).

2.1 Timestamp

We selected a curated set of five timestamps as a test-bed for the DEM models. These events are summarized here:

- **`20170910_1548`: 2017-09-10 15:48 UT** - AR 12673 / HARP 7115 (S08 W88, west limb).
  X8.2-class eruptive flaring event (M5.0+); *Region of Interest*: flare center at limb.
  References: [12].
- **`20110906_2217`: 2011-09-06 22:17 UT** - AR 11283 / HARP 833 (N14 W18, near disk center).
  X2.1-class eruptive flaring event (M5.0+); *Region of Interest*: flare center near disk center.
  References: [13].
- **`20140910_1731`: 2014-09-10 17:31 UT** - AR 12158 (N11 E05, near disk center).
  X1.6-class flare with filament eruption; *Region of Interest*: flare/filament core near disk center/ saturation area.
  References: [14].
- **`20120603_0000`: 2012-06-03 00:00 UT** - AR 11493 / HARP 1722 (disk center; coronal hole in FOV to the east).
  Quiet-Sun / coronal-hole context observation; *Region of Interest*: coronal hole near disk center.
  References: [15].
- **`20131113_0908`: 2013-11-13 09:08 UT** - AR 11890 (S11 W67).
  C2.4-class flaring event (event #2 in reference); *Region of Interest*: flare center on the west disk.
  References: [16].

[1] `https://triborough.cs.nyu.edu/vp2435/demdemo/webapp/flaresX.html`

2.2 Mode

We provide a set of visualizations summarized here:

- *Region of Interest*: DEM profiles of a selected 128x128 pixel extraction from the region of interest. From top row to bottom, a full disk overview is presented with the highlighted region, a zoom-in with AIA 193Å, and the reconstructed DEM profiles from logT 5.5 to 8.
- *Full Disk DEMs*: DEM profiles of the full disk. From top row to bottom, mean LogT, DEM profiles from logT 5.5 to 8.
- *Full Disk AIA*: AIA filtergrams of the full disk, all plotted with same min/max range. Left column: Synthesized AIA observation, right column: original AIA observation (make sure the 'Show Original AIA' box is checked).
- *Full Disk DEM+AIA*: DEM profiles in the first column with AIA synthesis in the second column.
- *Full Disk JPDFs*: Joint Probability Distribution Functions (JPDFs) of the full disk synthesis vs original. From top row to bottom, JPDFs of AIA 94, 131, 171, 193, 211, 335Å.
- *Pixel-wise DEMs*: DEM curves of specific pixels in 1) a bright region and 2) a quiet region. In the first row, a full disk AIA 193Å image is shown with a set of four different regions: "hot" (bright), quiet, flaring, coronal hole[2]. From row two onward, we show 50 different pixels' DEM curves for a bright and quiet area. From top to bottom, pixels are selected from darkest to brightest in the selected region. Each plot shows: 1) the original AIA intensity (blue) and resynthesized intensity (orange), 2) the inferred DEM curve (dark blue) with the 90% confidence interval (light blue).
- *Region JPDFs*: Joint Probability Distribution Functions (JPDFs) of the region of interest synthesis vs original. From top row to bottom, each region of interest as explained above. First column is the region overview with AIA 193Å, columns 2-7 show JPDFs of AIA 94, 131, 171, 193, 211, 335Å, as labeled.

3 Future Work

Having developed initial effective systems, we have several things to do that will be accomplished in the following months. Many of these will require input from the community (you!): (1) identify which particular version of the analytic DEM solver we will emulate; over the past few months, we have identified and fixed small issues in the Cheung et al. method (e.g., poor constraints on high-temperature bins, truncations in the solver's basis that make the inferred DEMs non-Gaussian). (2) investigate additional improvements, such as slight tweaks to improve AIA resynthesis quality and fine-tuning our handling of uncertainty; and (3) extensively validate the method and the uncertainty estimates, including comparisons across a range of conditions and explanations of *why* the method works.

[2]Take into account these were selected manually based on AIA 193Å. Some timestamps might lack one or two regions. For these we selected an additional hot/quiet region depending on the original target.

A APPENDIX: EXPLANATION OF REFERENCE DEM METHODS

Basis Pursuit Sparse DEM. The Basis Pursuit Sparse DEM method [6] finds the sparsest (in the ℓ_1 sense) solution that reconstructs the observation to within t standard deviations of the observation, or

$$\arg\min_{\mathbf{x}\geq 0} \mathbf{1}^\top \mathbf{x} \text{ s.t. } (\mathbf{o} - t\boldsymbol{\sigma}) \leq \mathbf{R}\mathbf{x} \leq (\mathbf{o} + t\boldsymbol{\sigma}). \tag{1}$$

Note that there are no constraints on the goodness of fit except for being within t standard deviations of the observation. Since $\mathbf{R}$ and $\mathbf{x}$ are both non-negative, if there is any uncertainty, then re-synthesized observation $\mathbf{Rx}$ is likely an underestimate: if $\mathbf{Rx}$ is an overestimate, then one can reduce the value of $\mathbf{x}$ and get a better objective. Making this argument more formally, of course depends on the structure of $\mathbf{R}$, but this is observed in practice. Additionally, there may *not* be a solution within the given tolerance due to noise or model mis-specification. Typically, one iterative relaxes the tightness, and re-fits (e.g., solving first with $t = 1.4$, then relaxing to $t = 2.8$, etc.).

Equation 1 is a linear program and can be solved efficiently by a number of methods (e.g., Simplex, Dual Simplex) that are implemented by a variety of packages.

Elastic Net DEM. Elastic Net [7] aims to solve a regularized reconstruction problem that has a data-fidelity/goodness-of-fit term that measures the distance between $\mathbf{o}$ and $\mathbf{R}$ as well as regularization terms including the ℓ_1 norm and the squared ℓ_2 norm. The formulation is

$$\arg\min_{\mathbf{x}\geq 0} \frac{1}{2C}||\mathbf{o} - \mathbf{R}\mathbf{x}||_2^2 + \alpha\lambda||\mathbf{x}||_1 + \frac{1}{2}\alpha(1-\lambda)||\mathbf{x}||_2^2, \tag{2}$$

where $\alpha \in \mathbb{R}^+$ is a regularization strength and $\lambda \in [0, 1]$ is a tradeoff parameter between ℓ_1 and ℓ_2 regularization. Lasso (as used in the MUSE SDC paper [3]) is a special case where $\lambda = 1$.

Note that there are no guarantees about the quality of the fit: the system continuously trades off between data fidelity and preferring a smaller solution $\mathbf{x}$. So $\mathbf{Rx}$ can be arbitrarily far from $\mathbf{o}$.

Equation 2 can be solved by standard solvers, including coordinate descent. Special cases of Equation 2 like Lasso can be solved with specialized solvers; for example, `LassoLars` estimator in Python's package `scikit-learn`.

B APPENDIX: DECONVOLUTION EXPERIMENTS

To account for the impact of AIA's point spread function (PSF) on DEM inversion quality, we conducted experiments comparing deconvolved and non-deconvolved AIA inputs. The AIA instrument has a well-characterized PSF that causes spatial blurring in the observed images due to optical effects including scattered light. Deconvolution aims to reverse this convolution effect, potentially recovering finer spatial structures in the derived DEM maps. We generated solutions by running the BP and ENet inversion methods on both deconvolved and non-deconvolved AIA observations. These are direct applications of the reference solvers (not neural network emulations), allowing us to isolate the impact of deconvolution on the optimization inversions themselves. For the deconvolution process, we used `aiapy`'s deconvolve function (`aiapy.psf.deconvolve`) with the default PSF calculation method (`aiapy.psf.calculate_psf`)[3]. This gives four cases to compare: BP with deconvolution, BP without deconvolution, ENet with deconvolution, and ENet without deconvolution.

We analyzed the results and found little difference in the overall quality of the DEM reconstructions. However, with the aim to obtain relevant comments from the reviewers, we release these results in a separate web visualizer[4] that allows for side-by-side comparisons across the test timestamps and visualization modes described in Section 2.

[3]https://aiapy.readthedocs.io/en/stable/generated/gallery/psf_deconvolution.html
[4]https://triborough.cs.nyu.edu/vp2435/demdemo/webapp/flaresX_deconv.html

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